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Unit 5 : Session 2 : SLO 2 | Memory Consistency and Communication Overhead

I. Multiple Choice

1. b) A contract that defines the order in which memory operations appear to execute.
2. b) Time spent transferring data between processors or nodes.
3. b) Memory coherence problem “inconsistent views of shared data across processors.
4. c) Parallel execution may become slower than sequential execution “overhead exceeds the benefit.
5. b) Sequential consistency “all processors observe all writes in exactly the same order.

II. True or False

1. TRUE “consistency models define a common observed order of memory operations.
2. FALSE “weak models deliberately relax immediate visibility for performance.
3. TRUE “frequent data exchange across the network raises communication overhead.
4. TRUE “lower communication overhead means adding processors keeps paying off (better scalability).
5. FALSE “coherence is about keeping cached copies identical; consistency is about the ordering of operations “related but distinct.

III. Scenario-Based Activity

- Issue 1 “inconsistent results across runs: a memory consistency/coherence problem. Processors exchange intermediate results but observe writes in different orders (relaxed visibility), so computations consume stale values; non-deterministic ordering makes each run differ.
- Issue 2 “execution time grows with more processors: communication overhead. With 64 nodes exchanging results frequently, message-passing time grows faster than the per-node computation shrinks, so total time increases “the classic scalability limit (overhead > computation).
- Fixes: enforce synchronization/consistency at exchange points (barriers, locks, release consistency), reduce communication frequency (larger messages, better partitioning), and overlap communication with computation.